

Reinforcement Learning for Thrust-Vectored Fin Control of an Electric Ducted Fan VTOL Vehicle in Simulation

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Abstract—This paper studies closed-loop control of a thrust-vector electric ducted fan (EDF) vertical take-off and landing (VTOL) vehicle whose only attitude actuators are four jet vanes immersed in the fan exhaust. Attitude authority is therefore coupled to the throttle state, and every change in rotor speed exchanges angular momentum with the airframe. We develop a GPU-accelerated simulation in NVIDIA Isaac Sim and, in this revision, correct its physics: a finite-momentum jet model replaces independent per-vane lift forces, the EDF is coupled to a LiPo equivalent circuit, the full rotor spool and gyroscopic reactions are integrated with an energy-conserving implicit-midpoint scheme, and external forces are applied about the true center of mass. On the corrected plant we compare model-free Proximal Policy Optimization (PPO) with a convex-optimization guidance controller that adapts the lossless-convexification powered-descent method of Açikmeşe and Ploen to a constant-mass, battery-powered vehicle. The planner minimizes electrical energy with a power-cone objective and adds a thrust-rate bound derived from the vanes’ yaw authority. It landed in all seven benchmark missions within the vehicle’s envelope, at touchdown speeds of 0.146–0.184 m/s and 0.05–0.40 m from the pad center. These missions included combined wind, sensor noise and center-of-mass offset, a 50 m cold-rotor start, and a two-waypoint route. The controller also flew a 100 s three-hover route, and it landed four further starts from about 100 m in wind with sensor noise, at initial speeds up to 30 m/s. A PID baseline touched down at 1.50 m/s on the default mission. On the same plant, the best PPO policy reached 1.3% success on its full randomized waypoint task after 700 million transitions. On this experiment, explicit model-based convex guidance substantially outperforms model-free PPO. We identify the rotor–body angular-momentum exchange as the dominant obstacle to learning, and discuss the transfer risks of both approaches.

Index Terms—reinforcement learning, convex optimization, powered descent guidance, lossless convexification, thrust vector control, electric ducted fan, proximal policy optimization, VTOL, Isaac Sim, jet vane

I. Introduction

Vertical take-off and landing (VTOL) vehicles have seen rapid adoption across commercial, military, and research domains. While conventional multirotors achieve attitude control through differential rotor speeds, alternative configurations—particularly those employing thrust vector control (TVC)—offer potential advantages in aerodynamic efficiency, noise reduction, and mechanical simplicity for certain mission profiles [1].

Thrust-vector vehicles that use aerodynamic control surfaces (jet vanes or fins) immersed in the exhaust stream of a single propulsion unit present a fundamentally different control challenge compared to multirotors. The control authority is coupled to the throttle state: at low thrust, fin effectiveness diminishes because the dynamic pressure in the exhaust drops quadratically with rotor speed. A single fast rotor also stores substantial angular momentum. Each throttle change therefore torques the airframe about the thrust axis, and each body rotation produces gyroscopic precession. This coupling, together with nonlinear vane aerodynamics and hobby-grade servo dynamics, makes classical linear control design challenging.

Reinforcement learning (RL) has demonstrated success in quadrotor control [2], [3], fixed-wing flight [4], and rocket landing [5]. Convex optimization has an equally strong record in powered-descent guidance, where lossless convexification turns a non-convex thrust constraint into a problem that can be solved to global optimality in real time [21], [22]. Neither approach has been systematically applied to thrust-vector fin control of EDF vehicles. The key distinctions of this control problem include: (1) a single thrust source with four coupled control surfaces, (2) vane forces bounded by the momentum of a finite jet, (3) significant actuator dynamics and deadband from hobby-grade servos, and (4) rotor–body angular-momentum exchange strong enough to exceed the vanes’ yaw authority.

This paper makes the following contributions:

- A GPU-accelerated Isaac Sim environment for an EDF TVC drone, with a corrected physics model: finite-momentum jet vanes, a battery-coupled EDF, full rotor angular-momentum reactions integrated without numerical energy gain, and force application about the true center of mass. The physical and numerical corrections are each backed by regression tests.
- A convex (conic) guidance and flight-control stack for this vehicle, adapted from lossless-convexification powered descent guidance [21]. It uses a first-order-hold thrust profile, a thrust-rate bound derived from rotor momentum exchange, a minimum-electrical-

energy objective and waypoint routes.

- PPO policies for hover, landing, and waypoint flight, together with an analysis of reward design for TVC vehicles, including the terminal-vs-per-step magnitude balance critical for landing tasks.
- A comparison on the corrected plant showing that the convex controller lands reliably where both PPO and PID fail, with an analysis of why.

II. Related Work

A. Reinforcement Learning for Aerial Vehicle Control

RL-based control of aerial vehicles has progressed significantly in recent years. Hwangbo et al. demonstrated that PPO could learn agile quadrotor control policies that transfer to hardware [2]. Koch et al. applied deep RL to quadrotor attitude control, comparing multiple algorithms and finding that PPO provided robust training stability [3]. Pi et al. showed that RL policies trained in simulation could achieve robust quadrotor flight under wind disturbances after sim-to-real transfer [4].

GPU-accelerated parallel simulation has been a key enabler. Rudin et al. demonstrated that massively parallel environments in Isaac Gym could train locomotion policies orders of magnitude faster than prior approaches [6]. Makoviychuk et al. extended this to manipulation tasks with thousands of parallel environments [7].

B. Transformer-Based Memory for Long-Horizon Control

Landing tasks are inherently long-horizon: the controller must plan a descent trajectory over hundreds of decision steps while inferring slowly varying disturbances from local state snapshots. Standard feed-forward policies lack temporal context beyond the current observation. Transformer-XL [17] addresses this through segment-level recurrence, and Gated Transformer-XL (GT-XL) [16] stabilizes transformer training in RL by gating the residual connections. Li et al. [18] survey the use of transformers in RL and argue that they are most beneficial when long-range temporal credit assignment matters more than raw simplicity.

In aerospace guidance, Federici et al. [19] applied meta-reinforcement learning with transformers to lunar landing, and Carradori et al. [20] demonstrated transformer-based robust feedback guidance for atmospheric powered landing. Both studies report strong simulated performance under uncertain initial conditions and disturbances, but neither closes the loop on hardware transfer—a gap that motivates the physical testbed described in this work.

C. Convex Optimization for Powered Descent Guidance

Açıkmeşe and Ploen posed Mars powered descent as a convex second-order cone program (SOCP) [21]. The thrust magnitude must lie in an annulus $\rho_1 \leq \|\mathbf{T}\| \leq \rho_2$, which is non-convex. They relaxed it with a slack variable and proved that the relaxation is lossless: the optimum of the relaxed problem satisfies the original

constraint. Açıkmeşe and Blackmore generalized lossless convexification to a class of optimal control problems with non-convex control constraints [22]. Blackmore et al. added a minimum-landing-error formulation for targets that cannot be reached [23]. Because interior-point conic solvers have deterministic convergence behavior, these problems can be solved onboard. We use the Clarabel interior-point solver [24].

D. Thrust Vector Control

TVC has a long history in rocketry, where gimbaled nozzles or jet vanes provide attitude control during powered flight [8]. More recently, TVC has been explored for VTOL vehicles. Jet-vane TVC, where flat plates are immersed in the exhaust stream, provides a mechanically simple alternative to gimbaled motors but introduces nonlinear aerodynamic coupling [9].

Classical control approaches for TVC vehicles typically linearize the plant around a hover operating point and apply PID or LQR controllers [10]. However, the strongly nonlinear relationship between fin deflection, throttle state, and resulting forces—particularly the quadratic dependence of fin effectiveness on exhaust velocity—limits the operating envelope of linear controllers. Geometric attitude control on $SO(3)$ [25] avoids the singularities of Euler-angle linearization and is used by our convex flight controller’s inner loop. Optimization-based guidance can plan fuel- or energy-aware trajectories that respect actuator limits explicitly. Sequential convex programming extends this to non-convex constraints, but its trust-region iterations can become infeasible when the required maneuver departs far from the linearization point.

E. Sim-to-Real Transfer

Domain randomization [11] and system identification [12] are established techniques for bridging the simulation-to-reality gap. For aerial vehicles, Sadeghi and Levine demonstrated that policies trained entirely in simulation could transfer to real quadrotors [13]. The fidelity of the simulation dynamics model is a critical factor, motivating the physics corrections reported in this work.

III. Vehicle Description

A. Physical Platform

The testbed is a custom-built EDF VTOL vehicle (Fig. 1) with the following specifications:

Two propulsion profiles are modeled. The legacy 6S profile is the originally specified FMS 90 mm 12-blade EDF (FMSDF010-1) with a 4068-KV1850 inrunner and a 6S LiPo. The planned 8S profile is the upgrade selected for the next hardware build: a 90 mm 12-blade EDF with a 4075-KV1500 motor on an 8S pack. It is used for all corrected-plant results. The 8S parts have not yet been purchased, so its parameters are planning estimates, not

²Not all structural components are shown; only the core EDF unit, servos, frame, battery, and legs are present in this rendering.



Fig. 1. CAD rendering of the EDF TVC drone testbed showing the single axial EDF unit, four cruciform jet-vane fins, MG996R servo actuators, carbon-fiber frame, battery, and landing legs.²

TABLE I
Vehicle Physical Parameters

Parameter	Value	Source
Body mass	3.1 kg	Measured
Vane mass (each)	0.001 kg	USD asset
Total mass	3.104 kg	USD asset
COM offset (above origin)	10 mm	Estimate
Body length	0.35 m	Measured
Body diameter	0.12 m	Measured
I_{xx}, I_{yy}	0.05 kg m ²	Isaac Sim MassAPI
I_{zz}	0.02 kg m ²	Isaac Sim MassAPI
Fin count	4	—
Fin area (each)	0.002 m ²	Measured
Vane COP radius	0.04 m	Estimate
Max fin deflection	15° (0.262 rad)	Measured

bench measurements. On 6S, the pack sags under load to a thrust-to-weight ratio of about 1.1, too little to arrest a fall from a cold-rotor start (Section VIII-B). Four MG996R hobby servos actuate flat-plate jet vanes positioned in the EDF exhaust stream.

B. Control Architecture

The four vanes are arranged in a cruciform pattern at 90° intervals. Each vane hinges about its own radial span (confirmed by the builder), so a deflection turns its slice of the jet tangentially:

- Forward fin (+X): hinge axis along +X.
- Right fin (+Y): hinge axis along +Y.
- Aft fin (−X): hinge axis along −X.
- Left fin (−Y): hinge axis along −Y.

The vane forces act about 0.1 m below the body origin and 0.04 m off the thrust axis. A differential deflection of the forward/aft pair therefore produces a roll moment, and one of the right/left pair produces a pitch moment. A common-mode deflection of all four vanes produces a yaw moment, because the four tangential forces add around the axis. The vanes also push the vehicle sideways directly. This side force is strong: identified at about 59 m/s² of lateral acceleration per radian of vane effort at full rotor

speed. Altitude is controlled via the EDF throttle. This architecture provides three-axis rotational control plus thrust through five actuator commands: four fin angles and one throttle. An earlier version of the simulation modeled the hinges tangentially, which gave roll and pitch authority but no yaw authority. Section IV-K lists this and the other corrections.

IV. Simulation Environment

A. Coordinate Frames

The simulation uses two coordinate conventions. The Isaac Sim world frame is right-handed and Z-up (+x forward, +y left, +z up). Internal control and aerodynamic calculations use a body-fixed forward-right-down (FRD) frame (+x forward, +y right, +z down). A single conversion boundary between these frames is enforced through:

$$\mathbf{R}_{I \leftarrow F} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad (1)$$

A body-FRD vector is mapped to world coordinates by $\mathbf{v}^W = \mathbf{R}(\mathbf{q}) \mathbf{R}_{I \leftarrow F} \mathbf{v}^F$, where $\mathbf{R}(\mathbf{q})$ is the rotation matrix induced by the vehicle quaternion $\mathbf{q} = [w, x, y, z]^T$. Vehicle orientation uses the quaternion representation throughout to avoid gimbal lock.

B. Overview

The simulation environment is built on NVIDIA Isaac Sim using the Isaac Lab framework (v2.3.2), which provides GPU-accelerated rigid-body physics via PhysX 5. The environment implements a Gymnasium-compatible interface with vectorized step, reset, and observation computation across parallel environments on a single GPU. Each vane is a separate articulation link on a revolute joint, driven by a joint drive that follows the servo model.

Controllers act at $\Delta t_{RL} = 33.3$ ms (30 Hz). Physics runs at a higher rate. The original experiments used 120 Hz (decimation 4). Current PPO training uses 480 Hz (decimation 16) with 2,048–8,192 environments. The single-vehicle missions used to evaluate the classical controllers use 120 Hz with the same corrected force and integration models. Actuator dynamics, propulsion, battery, and aerodynamic forces are updated at every physics substep.

C. Propulsion and Battery Model

The EDF is modeled with first-order spool dynamics toward a voltage-scaled speed target, and a quadratic thrust law:

$$\dot{\omega} = \frac{\omega_{\text{target}} - \omega}{\tau_{\text{motor}}}, \quad \omega_{\text{target}} = \text{clip}\left(d \frac{V_{\text{bus}}}{V_{\text{ref}}}, 0, 1\right) \omega_{\text{max}} \quad (2)$$

$$T = k_T \omega^2, \quad k_T = T_{\text{max}} / \omega_{\text{max}}^2 \quad (3)$$

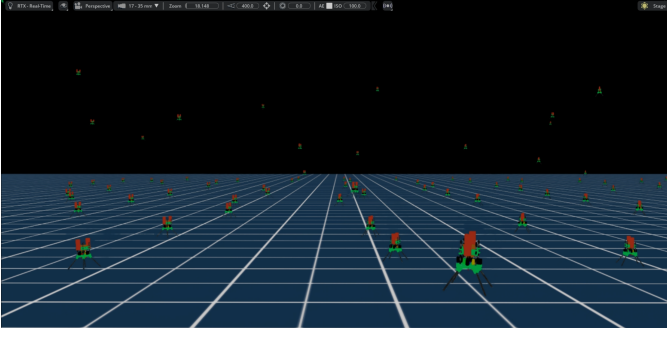


Fig. 2. NVIDIA Isaac Sim viewport showing 128 parallel EDF drone instances during a headed training run. Each vehicle is spawned at a randomized position and orientation within the configured spawn box and executes the landing task simultaneously on the GPU.

where ω is the rotor angular velocity, $d \in [0, 1]$ is the commanded duty (throttle), V_{bus} is the loaded pack voltage, and V_{ref} is the nominal pack voltage at which ω_{max} is defined. In steady state $T = T_{\text{max}}(dV_{\text{bus}}/V_{\text{ref}})^2$, so the same duty gives less thrust as the pack sags.

The aerodynamic shaft power follows the momentum-theory scaling $P_{\text{aero}} = P_{\text{max}}(\omega/\omega_{\text{max}})^3$. The electrical demand is

$$P_{\text{elec}} = \frac{P_{\text{aero}} + \frac{d}{dt} \left(\frac{1}{2} I_{\text{rotor}} \omega^2 \right)}{\eta_m} + P_{\text{aux}}, \quad (4)$$

with regenerative braking assumed to be zero. The pack is a Thevenin equivalent circuit: open-circuit voltage $E(\text{SOC})$ from a piecewise-linear LiPo discharge curve, series resistance, and one RC polarization branch. The load current is the high-voltage root of $P = I(E - V_p - IR)$, limited to 120 A. State of charge follows coulomb counting, and a lumped thermal state tracks pack temperature. The voltage-dependent load is solved by fixed-point iteration at every substep. When the pack cannot deliver the requested power, the rotor may gain only the kinetic energy that the delivered shaft power provides:

$$\omega_{k+1} \leq \sqrt{\omega_k^2 + \frac{2 \Delta t (P_{\text{shaft}} - P_{\text{aero}})}{I_{\text{rotor}}}}. \quad (5)$$

This bound is what makes cold-rotor starts from low altitude unrecoverable on this vehicle. Near full throttle the motor can add rotor energy at only about 3 kW, so thrust builds more slowly than a first-order model alone would predict.

On the 8S profile the vehicle hovers at a rotor fraction of about 0.84, drawing about 2.1 kW. This corresponds to a duty of 0.84 at the reference voltage, or about 0.77 on a full pack. The static thrust-to-weight ratio is 1.58, which falls as the pack sags and after vane losses are included.

D. Rotor Angular Momentum

The rotor's angular momentum $\mathbf{H} = I_{\text{rotor}} \omega \hat{\mathbf{e}}_z$ is not part of the PhysX articulation, so its reactions are applied to the body explicitly at full physical strength:

TABLE II
EDF and Battery Parameters (Planned 8S / Legacy 6S)

Parameter	Value	Source
Max thrust T_{max}	48.0 / 39.2 N	Est. / datasheet
Max rotor speed ω_{max}	4650 / 4300 rad/s	KV $\times V_{\text{ref}}$
Reference voltage V_{ref}	29.6 / 22.2 V	Nominal pack
Exhaust speed (nominal)	128 / 116 m/s	$\sqrt{2T/(\rho A)}$
Shaft power at ω_{max}	3072 / 2300 W	$T v_{\text{jet}}/2$
Motor time constant τ_{motor}	0.15 s	Estimate
Rotor inertia I_{rotor}	$2 \times 10^{-4} \text{ kg m}^2$	Estimate
Residual stator swirl fraction	0.1	Estimate
Motor efficiency η_m	0.88	Estimate
Pack capacity	5.0 A h	Planned
Cell resistance	3 m Ω	Estimate
Polarization R_p, τ_p	6 m Ω , 15 s	Estimate
Current limit	120 A	ESC continuous

- 1) Spool reaction: $\boldsymbol{\tau}_{\text{spool}} = -I_{\text{rotor}} \dot{\omega} \hat{\mathbf{e}}_z$, the reaction to rotor acceleration.
- 2) Gyroscopic precession: $\boldsymbol{\tau}_{\text{gyro}} = -\boldsymbol{\omega}_{\text{body}} \times \mathbf{H}$, coupling body rotation to rotor angular momentum.
- 3) Residual stator torque: $Q = f_{\text{res}} P_{\text{aero}}/\omega$, the steady shaft torque that the stator does not recover. The same Q leaves the nozzle as jet swirl, which the vanes see (Section IV-E).

The spool reaction dominates yaw. With $I_{\text{rotor}} \omega_{\text{max}}/I_{zz} = 2 \times 10^{-4} \cdot 4650/0.02 \approx 46.5 \text{ rad/s}$, every unit change in rotor fraction spins the body by about 46.5 rad/s about the thrust axis unless the vanes remove that momentum. A common-mode vane deflection provides only about $\pm 0.30 \text{ Nm}$ of yaw torque at hover. By contrast, a unit throttle step produces a transient spool torque of up to about 6.2 Nm. A throttle rate of 0.25/s uses 83% of the vanes' yaw authority. This single number shapes both the PPO action contract and the convex planner's thrust-rate constraint. At hover, $\|\mathbf{H}\| \approx 0.78 \text{ Nm s}$, which places a gyroscopic cross-coupling of about 16 rad/s between roll and pitch.

Integration. Applying $-\boldsymbol{\omega} \times \mathbf{H}$ at the old angular velocity is an explicit Euler step that adds rotational energy. In an unforced test at 240 Hz, it increased the body's rotational energy 10.2-fold in two seconds. PhysX's own explicit treatment of the rigid-body Euler term amplified the transverse rate-squared 114-fold at a yaw rate of 40 rad/s, even with the rotor stopped. We therefore solve the body Euler equation and the rotor term together with an implicit midpoint step,

$$\mathbf{I}(\boldsymbol{\omega}_m - \boldsymbol{\omega}_k) = \frac{\Delta t}{2} (\boldsymbol{\tau}_{\text{ext}} - \boldsymbol{\omega}_m \times (\mathbf{I} \boldsymbol{\omega}_m + \mathbf{H})), \quad (6)$$

with $\boldsymbol{\omega}_{k+1} = 2\boldsymbol{\omega}_m - \boldsymbol{\omega}_k$. In free flight this velocity step is paired with a Cayley orientation update $\mathbf{R}_{k+1} = \mathbf{R}_k \text{Cay}(\Delta t \boldsymbol{\omega}_m)$. The implementation also inverts PhysX's subsequent angular-momentum projection. The scheme introduces no damping and does not reduce the gyroscopic strength. In unforced two-second tests with roll and pitch rates of 6.28 rad/s and yaw rates up to $\pm 40 \text{ rad/s}$, the rotational-energy drift was at most 0.13% at 480 Hz and

0.12% at 960 Hz. The world angular-momentum error was below 1.2×10^{-3} N ms.

E. Jet-Vane Model

The original model applied an independent lift force $F_n = qSC_N(\alpha)$ to each vane, with dynamic pressure scaled by throttle. At 128 m/s, 0.002 m^2 and 15° , this could exceed 15 N of side force per vane. Yet a quarter of a 48 N jet carries only 12 N of axial momentum. The corrected model is a reduced-order control-volume model, following the momentum and angular-momentum bookkeeping of Drela's QPROP formulation [26]. It is not a blade-element fan model or CFD.

Four equal streamtubes share the mass flow $\dot{m} = T/U$, where U is the jet axial speed. U is reduced slightly so that axial and swirl kinetic power together equal the nominal wake power. The flow entering vane i is

$$\mathbf{v}_{\text{in},i} = U\hat{\mathbf{e}}_z + \Omega_s\hat{\mathbf{e}}_z \times \mathbf{r}_i + \boldsymbol{\omega}_{\text{body}} \times \mathbf{r}_{\text{exit}} - \dot{\mathbf{r}}_i, \quad (7)$$

where $\Omega_s = Q/(\dot{m}\langle r^2 \rangle)$ is the residual swirl rate, $\mathbf{r}_i$ is the vane's measured, moving center of pressure (COP), $\mathbf{r}_{\text{exit}}$ is the nozzle exit, and $\dot{\mathbf{r}}_i$ is the COP velocity relative to the translating body. A vane removes a bounded fraction of the flow normal to its actual plane $\hat{\mathbf{n}}_i$:

$$\lambda_i = \frac{qSk_{\text{duct}}C_{N_\alpha}}{\dot{m}_iU}, \quad \mathbf{v}' = \mathbf{v}_{\text{in}} - (1 - e^{-\lambda_i})(\mathbf{v}_{\text{in}} \cdot \hat{\mathbf{n}}_i)\hat{\mathbf{n}}_i, \quad (8)$$

with $q = \frac{1}{2}\rho U^2$ and $\dot{m}_i = \dot{m}/4$. Profile drag similarly removes a fraction $1 - e^{-\lambda_D}$, with λ_D built on C_{D_0} , of the streamwise component along the vane chord, giving $\mathbf{v}_{\text{out},i}$. The force on the vane is the momentum change of its streamtube,

$$\mathbf{F}_i = \dot{m}_i(\mathbf{v}_{\text{in},i} - \mathbf{v}_{\text{out},i}), \quad (9)$$

applied once, at the vane's actual COP. At small loading, (8) recovers the thin-airfoil slope $F_n \approx qSk_{\text{duct}}C_{N_\alpha}\alpha$ of the original model. At high loading the flow approaches alignment with the vane, instead of reversing more transverse momentum than its streamtube carries. Induced losses follow from the momentum turning and are not added a second time. Because the inflow includes swirl, body rotation and vane motion, the model is passive and provides flow damping through physical relative velocity rather than through a tuned damping torque. At zero deflection the vanes cancel the residual swirl torque: the net yaw torque is 0.000 N m from 0.5 to 1.0 throttle. The coefficients $C_{N_\alpha} = 3.5 \text{ rad}^{-1}$, $C_{D_0} = 0.05$ and $k_{\text{duct}} = 1.3$ are retained from the original semi-empirical model as estimates.

The quadratic coupling between throttle and fin effectiveness is preserved. At half rotor speed, the jet speed halves and the vane forces fall to about a quarter.

F. Force Application

Each vane force is applied at its measured, moving COP, and the EDF thrust is applied at the authored body origin,

so a center-of-mass offset produces a thrust-line moment. We found two defects in the position-aware wrench kernel of the installed Isaac Lab checkout. It crossed a world-frame lever arm with a force that had already been rotated into link coordinates. It also discarded a supplied torque whenever an application point was given. The environment therefore computes every moment about each body's actual center of mass itself and supplies force and torque without an application point. No vendor files were modified. The corrected application passes two Isaac regressions. First, 48 N of thrust with a ± 10 mm center-of-mass offset produces the expected $\pm 0.040 \text{ rad/s}$ pitch impulse; $\pm 0.0399 \text{ rad/s}$ was observed. Second, identical fin commands at four world headings produce the same body rates to within $1 \times 10^{-7} \text{ rad/s}$.

G. Servo Actuator Model

The MG996R servo is modeled as a first-order lag with rate limiting and a deadband on the tracking error:

$$\dot{\alpha}_{\text{fin}} = \text{clip}\left(\frac{e}{\tau_{\text{servo}}}, -\dot{\alpha}_{\text{max}}, \dot{\alpha}_{\text{max}}\right), \quad e = \begin{cases} \Delta & |\Delta| \geq b \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where $\Delta = \alpha_{\text{cmd}} - \alpha_{\text{fin}}$ and $\tau_{\text{servo}} = 0.05 \text{ s}$. The rate limit is $\dot{\alpha}_{\text{max}} = 6.98 \text{ rad/s}$, from the manufacturer's 0.15 s per 60° at 6 V; the original model assumed 0.14 s. The deadband is $b = 0.017 \text{ rad}$ ($\approx 1^\circ$), and the servo state is clamped to $\pm 0.262 \text{ rad}$. The servo target drives a PhysX joint drive whose torque is capped at the 1.08 N m stall torque. Under aerodynamic load the identified response is slower, with $\tau \approx 0.08 \text{ s}$ and a slew of about 2.5 rad/s.

The deadband on the error makes the servo stop b short of every command, so it behaves as backlash. Section V-B shows that this backlash sustains a real attitude limit cycle.

H. Body Drag

Translational drag is

$$\mathbf{F}_{\text{drag}} = -\frac{1}{2}\rho C_d A_{\text{ref}} |\mathbf{v}_{\text{rel}}|^2 \hat{\mathbf{v}}_{\text{rel}} \quad (11)$$

with $C_d = 1.0$ (bluff-body estimate) and $A_{\text{ref}} = 0.011 \text{ m}^2$. Drag is retained in calm air, and $\mathbf{v}_{\text{rel}}$ is the velocity relative to the wind field. The original simulation also applied a linear angular damper of 0.27 N m s/rad that had been tuned in simulation, with no physical source. It is replaced by the still-air cross-flow drag of a slender cylinder,

$$\boldsymbol{\tau}_{xy} = -\frac{\rho C_d D L^4}{64} \|\boldsymbol{\omega}_{xy}\| \boldsymbol{\omega}_{xy}, \quad (12)$$

which is small. Axial skin friction is omitted pending measurement.

I. Contact Detection and State Machine

Landing and crash detection use PhysX contact reports. The landing legs are collision shapes on the body link. A landing contact is a normal force of at least 1N on the body, and any contact on a vane link is unsafe. A four-state machine—AIRBORNE, GROUND_CONTACT_CANDIDATE, LANDED, CRASHED—requires 0.125s of continuous contact before LANDED, with the dwell counted in physics frames. A crash on the final dwell frame takes precedence over a new LANDED transition. The recorded impact speed is the maximum downward speed at any first contact, preserved across bounces, so a later gentle re-contact cannot hide a hard first impact. A soft landing requires an impact speed of at most 0.25m/s and a pad error of at most 0.5m; mission evaluation also requires the vehicle to stay settled after motor shutdown. The original simulation used a kinematic proxy instead (height below 0.40m and downward speed below 0.50m/s).

J. Disturbances

Disturbances can be enabled individually or combined, and a combination retains each source’s standalone strength. They are steady wind with stochastic gusts, which act through the relative airspeed in (11); sensor noise on the observation; and a center-of-mass shift of up to ± 1 cm per axis.

K. Physics Corrections and Verification

Table III summarizes the corrections made since the original experiments. Each is covered by a unit, pure-tensor or Isaac regression test; all 294 unit tests pass. These tests establish that the implementation is consistent with its stated parameters. They do not establish that those parameters match hardware.

V. Convex Optimization Guidance

The convex controller is an explicit classical controller, in the same class as the PID baseline. It is independent of PPO and is not used in any PPO training. It has two layers. A planner solves a conic program for a thrust trajectory from the measured state, through the mission’s waypoints, to a gate above the pad, and re-solves it in closed loop. A flight controller tracks the plan and converts thrust commands into attitude, vane and throttle commands.

A. Planning Problem

The vehicle is planned as a constant-mass point with thrust acceleration $\mathbf{u} = \mathbf{T}/m$ and gravity $\mathbf{g}$:

$$\ddot{\mathbf{r}} = \mathbf{u} + \mathbf{g}. \quad (13)$$

Because the battery vehicle does not lose mass, the logarithmic mass change of variables used for rockets in [21] is unnecessary. Time is discretized at nodes $k = 0, \dots, N$. Açıkmış and Ploen’s Problem 4 holds the thrust constant

TABLE III
Physics Corrections Since the Original Experiments

Original model	Correction
Tangential hinge axes; no yaw authority	Radial hinges; rank-3 roll/pitch/yaw authority
Vane side force followed the deflected jet	Force is the reaction to the jet’s momentum change
Independent qSC_N force per vane	Finite shared mass flow, (8)–(9)
Spool and gyro torques scaled by 0 and 0.1	Full scale, with residual stator swirl
Explicit gyro integration (energy gain)	Coupled implicit midpoint and Cayley update (6)
Tuned 0.27 N ms/rad angular damper	Cross-flow drag (12) and vane flow damping
Fixed COPs	COPs follow the measured vane-link poses
Isaac Lab position-aware wrench	Moments about the true COM, computed explicitly
No battery; fixed thrust per throttle	LiPo circuit, voltage-scaled spool, energy cap
Kinematic contact proxy; bounces reset impact speed	PhysX contact; first-impact speed preserved
Hover throttle assumed 0.88 (6S)	Derived from all link masses and net thrust: 0.934 (6S), 0.84 (8S), at the reference voltage

between nodes (zero-order hold). That steps the thrust direction at every node, which the rate-limited rotor and attitude loop cannot follow. We instead use a first-order hold, with $\mathbf{u}$ linear between nodes, and integrate the dynamics exactly:

$$\mathbf{r}_{k+1} = \mathbf{r}_k + \Delta t \mathbf{v}_k + \Delta t^2 \left(\frac{1}{3} \mathbf{u}_k + \frac{1}{6} \mathbf{u}_{k+1} + \frac{1}{2} \mathbf{g} \right), \quad (14)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \Delta t \left(\frac{1}{2} (\mathbf{u}_k + \mathbf{u}_{k+1}) + \mathbf{g} \right). \quad (15)$$

Lossless convexification. The thrust magnitude must satisfy $\rho_1 \leq \|\mathbf{T}\| \leq \rho_2$. The lower bound is positive because the vanes need jet flow for authority, so the feasible set is a non-convex annulus. Following [21], it is relaxed with a slack σ_k :

$$\|\mathbf{u}_k\| \leq \sigma_k, \quad \rho_1/m \leq \sigma_k \leq \rho_2/m. \quad (16)$$

By Lemma 1 of [21], $\|\mathbf{u}_k\| = \sigma_k$ at the optimum of a cost that increases with σ . Every plan reports the gap $\max_k(\sigma_k - \|\mathbf{u}_k\|)/\sigma_k$ as a numerical check. The upper bound keeps half of the thrust margin in reserve for the tracking loop, $\rho_2 = W + 0.5(T_{\text{avail}} - W)$, where T_{avail} is evaluated at the measured bus voltage. The lower bound is at least 60% of weight W for vane authority. It is raised further so that the planned downward acceleration never exceeds the planned braking acceleration: $W - \rho_1 \leq \rho_2 - W$.

Constraints. In addition to (14)–(16), and to the initial

state and the rotor's present thrust vector $\mathbf{u}_0$:

$$\hat{\mathbf{e}}_z^T \mathbf{u}_k \geq \sigma_k \cos \theta_p, \quad \theta_p = 15^\circ \quad (17)$$

$$\|\mathbf{u}_{k+1} - \mathbf{u}_k\| \leq \dot{u}_{\max} \Delta t_k \quad (18)$$

$$\|\mathbf{v}_k\| \leq V_{\max}, \quad V_{\max} = 4 \text{ m/s} \quad (19)$$

$$\|(\mathbf{r}_k - \mathbf{r}_{\text{td}})_{xy}\| \leq \tan \gamma_k (r_{k,z} - z_{\text{td}}), \quad \gamma = 45^\circ \quad (20)$$

$$\mathbf{r}_N = \mathbf{r}_{\text{gate}}, \quad \mathbf{v}_N = \mathbf{v}_{\text{gate}}, \quad \mathbf{u}_N = -\mathbf{g}. \quad (21)$$

Constraints (17), (19) and (20) are the pointing, speed and glide-slope constraints of [21]. The gate thrust in (21) is vertical and equal to weight, so the hand-over to terminal descent has no thrust step. The other constraints are specific to this vehicle:

- Thrust-vector rate (18). A rotor speed change yaws the body (Section IV-D). The bound is the same 0.25/s duty rate used by the PPO action contract, converted to thrust at the lowest planned throttle. It bounds the vector rather than the slack; with a slack-rate bound, the lossless gap reached 3.4% and the executed thrust stepped faster than the rotor could follow. With the first-order hold, the bound limits the continuous thrust and also caps the thrust-direction slew that the attitude loop must follow.
- Glide cone taper. A start outside the 45° cone gets a widened cone, which narrows back to 45° three seconds before the gate. The landing-leg nodes must also stay at or above the gate altitude.
- Speed braking. A start faster than $V_{\max}$ keeps its speed and then brakes back under the bound, instead of being declared infeasible.
- Waypoints. A fly-through node must lie within half the capture radius. A hover node is at rest on the point, with thrust equal to weight, for the remaining hold time.

Objective. The planner minimizes electrical energy. The simulator's power law $P = P_{\text{ref}}(T/T_{\text{ref}})^{3/2} + P_{\text{aux}}$ is convex in T . It enters the problem through a power-cone constraint $c_k \geq \sigma_k^{3/2}$ at each node, integrated with the trapezoid rule:

$$\min \sum_{k=0}^N w_k c_k \quad \text{s.t.} \quad (c_k, 1, \sigma_k) \in \mathcal{K}_{\text{pow}}^{2/3}. \quad (22)$$

The alternative objective, the integral of σ , is the propulsive- Δv analog of the paper's fuel cost; with it, the problem is a pure SOCP.

Free final time and infeasible starts. The landing-leg duration is chosen by a golden-section line search, following Algorithm 1 of [21]. Re-plans search around the previous plan's remaining time. When no duration is feasible (for example, too fast and too low to stop), a soft-terminal problem minimizes the miss distance and heavily penalizes ground penetration, in the spirit of [23]. It keeps every actuator constraint, so it degrades to maximum braking.

The conic program is solved with Clarabel 0.11.1 [24]. A single solve takes about 3 ms. A complete plan, comprising 10–30 solves of the line search, takes 30 ms to 110 ms on the host CPU without warm starting.

B. Flight Controller

The flight controller reads only the observation (including any sensor noise), the mission sequencer's remaining route, and the measured bus voltage. It runs through four phases:

- 1) Spool-up (cold rotor): attitude hold while duty ramps at 4/s. This trades a yaw transient for altitude lost while falling.
- 2) Powered descent / route: track the plan, re-solved every 0.5 s. While the tracking error is below 0.75 m and 0.75 m/s, a new plan starts from the current plan's reference state; otherwise it starts from the measured state. Re-planning stops in the last 0.8 s before the gate.
- 3) Terminal descent: from a gate 0.5 m above touch-down height, a velocity-commanded vertical descent at 0.15 m/s, clamped at 0.18 m/s. The descent pauses while the vehicle is off center, except below a 0.15 m commit height.
- 4) Touchdown: once a contact candidate appears, the attitude is held level and the rotor is unloaded to 50% of weight so that the contact dwell can complete.

The inner loops are as follows. Tracking commands $\mathbf{a}_{\text{cmd}} = \mathbf{u}_{\text{plan}}(t + 0.15 \text{ s}) + \text{PID}(\mathbf{r}_{\text{ref}} - \mathbf{r}, \mathbf{v}_{\text{ref}} - \mathbf{v})$. The plan's thrust is fed forward one spool lag ahead. The feedback is bounded to 3 m/s^2 per axis group, and conditional integration limits each integrator's authority to 1.5 m/s^2 . Attitude control uses the geometric error on $SO(3)$ [25]. It keeps the PID baseline's fin-effort structure, mixer and fixed rotor gyroscopic cross-feed, and gives roll and pitch priority over yaw. Yaw is rate-damped only, because heading does not matter for this axisymmetric vehicle. Throttle inverts $T = T_{\max}(d V_{\text{bus}}/V_{\text{ref}})^2$ at the measured bus voltage.

Plant identification and the deadband limit cycle. The gains were set from models identified on recorded Isaac missions ($R^2 \geq 0.99$; Table IV). The first version of the controller showed a 1.0–1.1 Hz retrograde coning in every recorded mission, with 3° of tilt and roll/pitch rates of $15^\circ/\text{s}$ RMS. Through the vane side force, the coning drove $\pm 0.5 \text{ m/s}$ of lateral velocity. Its cause is the servo deadband acting as backlash. With the rotor's angular momentum, the attitude loop's retrograde precession mode sits at about 176° of loop phase, so the loop is only conditionally stable, and the gain and phase removed by backlash sustain a limit cycle. An offline replica reproduced the cycle ($14.1^\circ/\text{s}$, 3.0° , 1.0 Hz), and

no gain set removed it. A deadband inverse, driven by the measured vane angle,

$$\alpha_{\text{cmd}} = \alpha_d + b \text{sat}\left(\frac{\alpha_d - \alpha_{\text{meas}}}{0.25b}\right), \quad (23)$$

reduced the coning to $1.2^\circ/\text{s}$ in the replica. In Isaac it reduced whole-flight roll/pitch RMS from $14.1^\circ/\text{s}$ to $1.7^\circ/\text{s}$ on the default mission. Over-compensation (a true deadband of zero) only dithers, at $1.7^\circ/\text{s}$. Under-compensation leaves proportionally more coning, so the real deadband must be measured on the bench.

TABLE IV
Identified Plant Models Used to Set the Convex Controller Gains

Channel	Model	Consequence
Yaw	$I_{zz}\dot{r} = -0.265r + 15.0f^2\delta_{cm}$	Yaw damper gain 0.03 (gain margin ≈ 2.4)
Roll/pitch	$I\dot{p} = 20.7f^2e_{\text{roll}} - 0.33p - 1.05Hq$	Gyro coupling $\approx 16 \text{ rad/s}$; fixed cross-feed 0.03
Lateral	59 m/s^2 per rad of vane effort	Lateral gains 0.5/0.8/0.15
Vanes	$\tau \approx 0.08 \text{ s}$, slew $\approx 2.5 \text{ rad/s}$, 1° backlash	$k_{d,\text{att}} = 0.08$; deadband inverse (23)

VI. Learning Algorithm

A. Proximal Policy Optimization

We employ Proximal Policy Optimization (PPO) [14] with clipped surrogate objectives. PPO is an on-policy actor-critic algorithm that constrains policy updates to a trust region, preventing destructively large gradient steps while maintaining sample efficiency. The clipped objective is:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right] \quad (24)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio, $\hat{A}_t$ is the Generalized Advantage Estimate (GAE) [15], and $\epsilon = 0.2$ is the clipping coefficient. An adaptive KL-divergence early stopping criterion ($\text{KL}_{\text{target}} = 0.03$) provides an additional safeguard against policy collapse.

Advantages are computed using GAE with $\lambda = 0.95$ and discount factor γ :

$$\hat{A}_t = \sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \quad (25)$$

The combined loss function optimized at each update is:

$$L(\theta) = L^{\text{CLIP}}(\theta) - c_1 L^{\text{VF}}(\theta) + c_2 H[\pi_\theta] \quad (26)$$

where L^{VF} is the value function mean-squared error loss ($c_1 = 0.5$), and $H[\pi_\theta]$ is the policy entropy bonus

that encourages exploration. Actions are tanh-squashed Gaussians. The corrected implementation computes the entropy and likelihood of the transformed distribution with its Jacobian, stores the pre-squash samples so that saturated actions keep exact likelihoods, clips the actor and critic gradients separately, and does not clip the value loss by default.

B. Observation Space

In the original landing experiments, the agent receives a 24-dimensional observation vector at each decision step:

TABLE V
Observation Space (Original Landing Task)

Index	Component	Dim
0–2	Position error (m)	3
3–6	Attitude quaternion $[w, x, y, z]$	4
7–9	Linear velocity, body FRD (m/s)	3
10–12	Angular velocity, body FRD (rad/s)	3
13	Height above ground (m)	1
14–17	Fin angles (rad)	4
18–21	Fin angular rates (rad/s)	4
22	Motor RPM, normalized $[0, 1]$	1
23	Contact state (int $\rightarrow$ float)	1

Position error is computed as the vector from the current position to the task target (hover setpoint or landing pad center). The quaternion representation avoids gimbal lock singularities inherent in Euler angles. Fin angles and rates provide proprioceptive feedback on actuator state, which is critical given the first-order servo lag. The normalized motor RPM informs the policy of available fin authority, since fin effectiveness scales quadratically with rotor speed. With the battery model, four electrical channels are appended (state of charge, normalized loaded voltage, current, and polarization voltage), giving 28 inputs.

C. Action Space

The action space is a 5-dimensional continuous vector:

$$\mathbf{a} = [\alpha_1, \alpha_2, \alpha_3, \alpha_4, u_{\text{throttle}}] \quad (27)$$

where $\alpha_i \in [-0.262, +0.262] \text{ rad}$ ($\pm 15^\circ$) are the four fin deflection commands and $u_{\text{throttle}} \in [0, 1]$ is the normalized throttle. The network outputs are in $(-1, 1)$ via tanh saturation; fin channels are linearly scaled to $\pm 0.262 \text{ rad}$ and the throttle channel is mapped via $(z + 1)/2$.

D. Network Architecture

Both the actor (policy) and critic (value function) are implemented as separate two-layer fully connected networks with 256 hidden units per layer and tanh activations (Fig. 3). The actor outputs 5 action means; the exploration noise is parameterized by a learnable per-dimension log-standard-deviation vector. The critic outputs a single scalar state value.

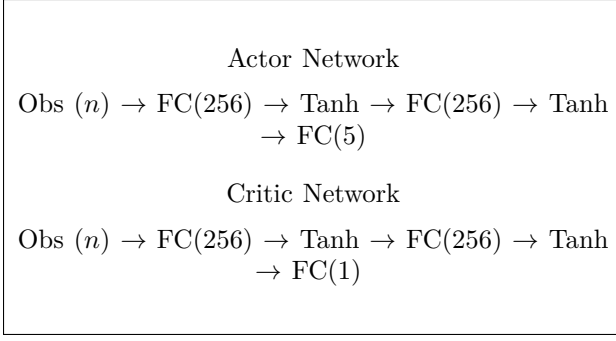


Fig. 3. Actor-critic network architecture. The observation width is $n = 24$ in the original landing task, 28 with battery channels, and 54 in the waypoint-flight task. The actor produces 5 action means; per-dimension log-standard-deviations are learned as free parameters.

Two initialization choices are critical for stable early training:

- 1) Throttle bias initialization. The actor output layer bias for the throttle channel is initialized to $\text{atanh}(2 \cdot 0.78 - 1) \approx 1.05$, so the initial mean throttle maps to ≈ 0.78 . This is below hover, so initial rollouts straddle the productive hover/descent basin rather than free-falling from the default 0.5 throttle. The original paper estimated hover at $\sqrt{mg/T_{\max}} \approx 0.88$. Once all link masses and vane drag are included, level hover on the 6S plant requires 0.934.
- 2) Per-channel exploration noise. Fin channels are initialized with $\log \sigma = -2.0$ (std ≈ 0.14), while the throttle channel uses $\log \sigma = -1.0$ (std ≈ 0.37). This asymmetry prevents excessive lateral noise from the fins while maintaining sufficient throttle exploration for altitude control.

E. Training Procedure

The original experiments used 128 parallel environments on a single NVIDIA GPU. Each PPO update collects a rollout of 128 steps per environment (16,384 transitions total), then performs 4 epochs of minibatch gradient descent with 8 minibatches per epoch. The optimizer is Adam with learning rate 3×10^{-4} and gradient norm clipping at 0.5. Table VI also lists the settings of the corrected-plant waypoint-flight task (Section VI-G).

For the hover task, training runs for 2,048 environment steps with a residual-PID architecture: a PID baseline controller provides the nominal action, and the PPO policy learns a bounded residual correction scaled by 0.05. The hover task uses a smaller rollout length (64 steps), 2 minibatches, 2 update epochs, and a reduced learning rate (10^{-6}) to avoid destabilizing the PID baseline. For the landing task, training runs for 5,000,000 environment steps with pure PPO (no PID baseline), as the landing maneuver requires the policy to learn the full descent-to-touchdown trajectory.

TABLE VI
PPO Hyperparameters

Parameter	Landing	Waypoint flight
Parallel environments	128	8,192
Rollout length	128 steps	32 steps
Minibatches per update	8	8
Update epochs	4	4
Learning rate	3×10^{-4}	KL-adaptive
Discount γ	0.99	0.999
GAE λ	0.95	0.95
Clip coefficient ϵ	0.2	0.2
Entropy coefficient	0.005	0.002
Value loss coefficient	0.5	0.5
Max gradient norm	0.5	0.5
Target KL	0.03	0.03
Value normalization	None	PopArt
Initial fin / throttle $\log \sigma$	-2.0 / -1.0	-2.5 / -1.0

F. Spawn-Position Curriculum

The landing task employs a linear spawn-position curriculum to address the exploration challenge of learning precision landing from large initial offsets. Training begins with a narrow spawn box of ± 0.5 m in the horizontal plane (XY), which linearly anneals to the full ± 2.0 m evaluation range over the first 3,000,000 steps. The altitude spawn range (8–12 m) remains constant throughout. This curriculum allows the policy to first learn on-pad touchdown behavior before generalizing to larger lateral offsets.

Evaluation is always conducted against the full (un-curricularized) spawn range to ensure reported metrics reflect the final task difficulty.

G. Corrected-Plant PPO: Waypoint Flight

On the corrected plant, landing-first PPO did not learn full-task landing (Section VIII-C). PPO was therefore reformulated as a single skill: reach the next waypoint efficiently. A mission is a sequence of waypoints. A fly-through is captured when the swept path passes within its radius. A hover requires a continuous hold inside its radius at no more than 0.5 m/s and $60^\circ/\text{s}$ body rate. A land waypoint, allowed only last, requires a real LANDED contact within the pad radius at no more than 0.25 m/s. The 54-channel body-frame observation contains the vectors to the active waypoint and the next two legs, the waypoint kinds, the capture radius and remaining hold time, the gravity direction, velocity, body rates, height, vane angles and rates, rotor speed, contact state, battery state, the previous action, and a fine-scale copy of the target vector.

Two explicit interface choices follow from the rotor momentum analysis of Section IV-D; both are part of the task definition. First, the policy commands a throttle rate, $d \leftarrow d + \tanh(z) \cdot 0.25 \Delta t_{\text{RL}}$. This bounds the spool reaction to 83% of the vanes' yaw authority and makes throttle exploration a random walk with lasting altitude effects. With direct duty commands, no run learned altitude feedback. Second, a fixed yaw-rate damper removes the

common mode of the policy’s vane commands and adds $-0.15r$ to every vane. The policy keeps the differential (roll and pitch) part. Before this damper was introduced, every policy developed a common-mode vane bias and orbited its targets.

The reward is energy-weighted. Per second of level hover, time costs 0.25 and energy costs 1.3 per Wh, or about 0.75, so energy is 75% of the efficiency cost. Dense guidance is potential-based [27], with $\Phi = -(\text{remaining route length})$ and $\Phi = 0$ at absorbing states. A yaw-weighted body-rate cost uses soft limits of $180/180/90^\circ/\text{s}$. Mission success earns +400 and every failure -400 . The nominal worst-case integrated step cost of a 90s full-power episode is 274, below the terminal magnitude. A 16-stage curriculum changes only the spawn, the mission generator and the episode length. The final task has 1–6 waypoints over 2–25 m legs at 2–30 m altitude, state of charge 50–100%, and steady wind up to 4 m/s. Training spawns the rotor at hover speed.

VII. Reward Design

Reward shaping is a critical design decision for TVC landing, where the agent must simultaneously manage altitude, lateral position, attitude, and touchdown conditions. We structure the reward as a sum of weighted per-step shaping terms and one-time terminal events. This section describes the original landing task.

A. Per-Step Shaping Terms

The per-step reward at each decision step is:

$$r_{\text{step}} = \sum_i w_i \cdot \phi_i(s_t, a_t) \quad (28)$$

where ϕ_i are individual reward terms and w_i are their weights. Table VII lists all terms and their weights for the landing task.

TABLE VII
Landing Task Reward Weights

Term	Weight
Per-step shaping	
Alive bonus	+0.10
3D position error	−0.05
Horizontal position error	−0.30
Attitude error (tilt)	−0.20
Angular velocity	−0.10
Control effort (fin $ \alpha $)	−0.02
Control rate (fin $ \dot{\alpha} $)	−0.01
Vertical speed shaping	−0.30
Delta-v cost $(T/T_{\text{max}})^2$	−0.20
Terminal events	
Crash penalty	−200
Touchdown softness	$+25 \cdot e^{-v_{\text{impact}}}$
Pad accuracy	$+200 \cdot e^{-d_{\text{horiz}}}$
Landing success	+250

B. Terminal-vs-Per-Step Magnitude Balance

A key insight for landing reward design is the magnitude balance between integrated per-step costs and one-time terminal rewards. For an episode of length T steps, the integrated per-step contribution is $\sum_i |w_i| \cdot \bar{\phi}_i \cdot T$, which can easily reach $\mathcal{O}(100)$ over a 30-second episode at 30 Hz ($T = 900$). If terminal rewards are not substantially larger, PPO will optimize per-step cost minimization—typically by reducing throttle to zero—rather than pursuing the terminal landing event.

We set terminal magnitudes ($|w_{\text{terminal}}| \geq 200$) to clearly dominate the integrated per-step budget. The landing success bonus of +250 plus the pad accuracy bonus (up to +200) provide a combined terminal reward of up to +450 for a perfect landing, which exceeds the worst-case integrated per-step cost. The crash penalty of -200 ensures that uncontrolled impacts are strongly discouraged. The later audit found a loophole in this table: a hard landing could still collect the partial softness and accuracy bonuses. The corrected landing task adds a -250 hard-landing penalty, which exceeds the maximum partial payout of 225.

C. Horizontal Guidance Shaping

The horizontal position error term ($w = -0.30$) provides a dense gradient toward the landing pad throughout the descent. This term is independent of altitude, so the agent receives lateral guidance even at high altitudes where the pad accuracy terminal bonus (which uses $e^{-d_{\text{horiz}}}$) has negligible gradient. The vertical speed shaping term ($w = -0.30$) penalizes deviation from a target descent rate of approximately 0.5 m/s, encouraging a controlled descent profile rather than a ballistic drop.

VIII. Experimental Results

We report two sets of results. Section VIII-A gives the original PPO results on the original plant. Sections VIII-B–VIII-D give results on the corrected plant: the convex controller, PPO, and their comparison.

A. Original-Plant PPO Results

These results were obtained before the corrections of Table III. That plant had tangential hinges, a reversed vane side-force sign, suppressed rotor momentum, a tuned angular damper, and a kinematic contact proxy that could register a landing after a bounce. They are kept for the record. They are not evidence of landing performance on the corrected plant. All experiments use nominal (zero-wind) conditions, with metrics computed over 128 parallel episodes with the full spawn range.

Hover. The hover policy uses a residual-PID architecture where a PID controller provides the baseline action and the PPO policy learns a small bounded correction (scale = 0.05). After 2,048 environment steps of training, the policy achieves:

TABLE VIII
Hover Stabilization Results (Original Plant)

Metric	Value
Mean position error	0.085 m
Max position error	0.215 m
Mean tilt	<0.001 rad
Max tilt	<0.001 rad
Mean angular rate	1.1×10^{-5} rad/s
Evaluation passed	Yes

The near-zero tilt and angular rate values reflect the PID baseline’s attitude stabilization and, in part, the tuned angular damper and suppressed rotor torques of the original plant.

Landing. The landing policy is trained with pure PPO (no PID baseline) for 5,000,000 environment steps with spawn-position curriculum. Table IX summarizes the final evaluation metrics.

TABLE IX
Landing Task Final Evaluation Results (Original Plant)

Metric	Value
Landing rate (kinematic proxy)	100%
Mean touchdown speed	0.126 m/s
Max touchdown speed	0.347 m/s
Mean pad distance	1.89 m
Max pad distance	4.56 m
Mean throttle (eval)	0.759
Max downward speed	9.75 m/s
On-pad fraction ($d < 0.5$ m)	2%

Under the kinematic proxy, the policy registered a landing in every episode. However, only 2% of episodes ended on the pad, with a mean pad distance of 1.89 m. The proxy’s touchdown speed was measured after any bounce, so it may understate the first impact.

Fig. 4 shows the mean episode reward over the course of training. The reward rises from ≈ -3 in the first million steps to consistently positive values after 2M steps, reaching $\approx +1.75$ by 5M steps. Negative reward means that integrated per-step costs dominate the return; positive reward means that terminal bonuses outweigh them.

B. Convex Guidance on the Corrected Plant

Each mission is flown once in Isaac Sim on the planned 8S profile with a full LiPo, unless noted, and succeeds only on a LANDED contact that stays settled after motor shutdown. The missions cover four conditions. A cold start has the rotor stopped at $t = 0$, so the vehicle falls while spooling up; a warm start has the rotor at hover speed. The combined disturbance applies wind with gusts, sensor noise, and a center-of-mass shift together. Table X lists the benchmark missions with the final controller.

The controller landed all seven benchmark missions within the vehicle’s envelope. Touchdown speeds were

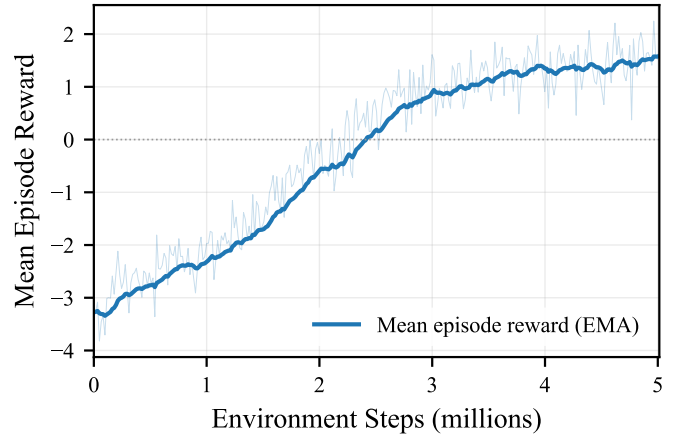


Fig. 4. Mean episode reward during original-plant landing training over 5M environment steps (EMA, $\alpha = 0.08$). Reward crosses from negative (dominated by per-step costs) to positive (dominated by terminal bonuses) around 2M steps.

0.146–0.184 m/s (gate 0.25 m/s) and pad errors 0.05–0.40 m (gate 0.5 m). The whole-flight roll/pitch rates were 1.5–13.3°/s RMS. The 50 m start obeys the 4 m/s speed bound. Its remaining 5.9 m/s peak is the fall during cold spool-up, before the first plan. The long route flew three hover waypoints with 10 s holds and then landed after 99.8 s, 0.16 m from the pad; its roll/pitch RMS of 3.9°/s compares with 15.8°/s before the deadband inverse. After the benchmark, four further missions from 100 m in wind with sensor noise all landed. Three started descending at 20 m/s, two of them with a total speed of 30 m/s. The fourth started 86 m from the pad horizontally, descending at 5.5 m/s, and flew one waypoint. Their impacts were 0.156–0.187 m/s and their pad errors 0.07–0.28 m, in 15–49 s. A fifth such mission was cancelled by the operator and is excluded.

The two failures lie outside the vehicle’s physical envelope rather than the controller’s. In the cold-rotor combined case, full thrust is commanded by $t = 0.5$ s. However, the energy cap (5) lets the motor add rotor energy at only about 3 kW. Thrust therefore reaches 40 N at 3.5 m altitude, with the vehicle still falling at 6.8 m/s, which needs more than 5 m to stop. On the legacy 6S pack, sag reduces the thrust-to-weight ratio to about 1.1, which cannot arrest the fall built up during a cold spool-up.

C. PPO on the Corrected Plant

Table XI summarizes PPO’s progress as the plant was corrected. Every entry is a direct-action feed-forward policy with no PID wrapper, thrust floor or scripted guidance.

The 30M-step policy on the intermediate plant landed reliably. That plant still applied independent per-vane forces that could exceed the jet’s momentum and a tuned angular damper. Once those were replaced by the finite-momentum jet and physical damping, full-task success

TABLE X
Convex Guidance: Benchmark Missions on the Corrected Plant (Final Controller)

Mission	Start (altitude, velocity, rotor)	Outcome	Impact (m/s)	Pad err. (m)	Time (s)	Energy (Wh)	Roll/pitch RMS ($^{\circ}$ /s)
Default	18 m, -1 m/s, cold	Landed	0.146	0.08	8.3	5.37	1.7
Offset + wind/gusts	12 m, 7.2 m off, -2 m/s, cold	Landed	0.152	0.30	8.3	5.58	13.3
Hot hop	3 m, 1 m off, warm	Landed	0.172	0.10	6.1	3.48	7.4
Combined disturbances	15 m, -5 m/s, warm	Landed	0.153	0.07	20.1	12.12	11.4
Route (fly-through + 2 s hover)	8 m, warm	Landed, 2/2 captured	0.150	0.40	23.5	13.82	8.3
High start	50 m, -1 m/s, cold	Landed	0.146	0.05	16.1	9.90	1.5
Legacy 6S	8 m, warm (0.9)	Landed	0.184	0.36	9.6	5.72	8.9
Long route (3 hovers, 10 s holds)	50 m, -0.5 m/s, warm	Landed, 3/3 captured	0.189	0.16	99.8	58.46	3.9
Combined disturbances	15 m, -5 m/s, cold	Crashed	—	—	—	—	—
Legacy 6S default	18 m, -1 m/s, cold	Hard landing	2.56	0.38	11.7	7.34	—

TABLE XI
PPO Across Plant Revisions (Independent Full-Task Evaluations)

Run	Plant	Transitions	Full-task result
8S landing, radial hinges	Independent vane forces, tuned damper	30.0M	95.7% success, 0.4% crash; 0.175 m/s, 0.13 m (superseded plant)
8S landing, momentum model	Finite-momentum jet, no damper	100.1M	0% success, 100% crash (16–20 m, cold rotor)
Landing-first waypoint missions	+ wrench and gyro-integration fixes	200M	0% in every evaluation to 132M; stalled on stage 1 of 11
Waypoint flight (Sec. VI-G)	Current plant, 480 Hz	700M	1.3% success (hover missions 2.1%, land 0%); 11.0% crash

fell to zero and did not recover within 100M transitions. The waypoint-flight reformulation learned hover and short hops. At 700M transitions, on 8,192 randomized full-task episodes, the outcomes were: 1.3% success, 51.4% geofence exits, 21.3% timeouts, 11.0% crashes, 7.5% hard landings, 5.0% premature landings, and 2.5% spin terminations above 360° /s. The 95th-percentile peak yaw rate was 328° /s. Successful episodes averaged 17.1 Wh over 29.0 s at a route speed of 0.69 m/s. At 750M transitions the policy had passed the landing stage at a relaxed 1.0 m/s touchdown gate (49% land-mission success). It had 0% success at the 0.25 m/s gate after a further 36M transitions.

The same 750M-step policy was flown through mission control on two three-waypoint trials. From a 50 m start, it was stopped at $t = 0.03$ s because the start lies above its 30 m training geofence. From a 32.5 m start, it did not capture the first waypoint in 100 s. Both starts are at or beyond the edge of its training envelope. The randomized evaluation above is therefore the fair measure of this policy.

D. Comparison

Table XII compares the three controllers on the corrected plant. The PID baseline flew the default mission with the same plant and battery and touched down at 1.50 m/s, six times the soft-landing gate.

On this experiment, the convex controller is clearly better. It lands softly and on the pad from every in-envelope start tested, including combined disturbances and routes, whereas PPO has not learned to land softly on the corrected plant. The comparison has limits. The convex controller was evaluated on 12 hand-specified missions, each flown once, whereas PPO was evaluated on thousands of randomized episodes. The convex controller has not yet been run on PPO’s randomized waypoint task.

TABLE XII
Controllers on the Corrected 8S Plant

	Convex	PID	PPO
Default mission (18 m, cold)	Landed, 0.146 m/s, 0.08 m	1.50 m/s touchdown	Outside envelope (cold rotor)
Benchmark landings	7/7	—	—
Waypoint routes	2/2 landed	Not supported	1.3% full-task success
Soft landing (0.25 m/s gate)	Yes, 0.146–0.189 m/s	No	Not yet learned
Cold-rotor starts	Landed when recoverable	1.50 m/s touchdown	Not trained
Training cost	None	None	700M transitions
Online cost	30–110 ms per plan	Negligible	One MLP pass

Running it there is the next step toward a like-for-like success rate.

IX. Discussion

A. Why Convex Guidance Outperforms PPO Here

The convex controller is given the physics that PPO must discover. Its thrust-rate constraint (18) encodes the rotor momentum exchange as a hard limit, and its energy objective (22) is the simulator’s own power law. The planner’s decisions are globally optimal for the point-mass problem at every re-plan, so it never has to explore to find a braking burn or a feasible descent profile. The inner loops were designed from identified plant models (Table IV).

PPO faces the same physics through exploration, and the corrected plant makes exploration costly in three ways. First, rotor momentum exchange couples throttle exploration to yaw. Throttle noise large enough to reveal altitude dynamics spins the body past its termination limit, while noise small enough to avoid that moves the

vertical speed by only about 0.02 m/s per step, too little for PPO to learn from. The rate-throttle contract and yaw damper were introduced for this reason. Second, soft landing requires sub-0.25 m/s vertical precision after a long descent. This is a sparse, delayed event that the policy reached at a 1.0 m/s gate but not at 0.25 m/s. Third, the persistent failure mode was climbing: extra throttle avoids early sinking crashes, but it drives the vehicle out of the geofence and raises rotor speed, which yaws the body. Throughout, the policy’s exploration noise collapsed before translation was learned.

The comparison also shows that the landing results of PPO depended strongly on the plant. An intermediate plant, with unbounded vane forces and an artificial damper, was much easier to control, and PPO reached 95.7% success on it. The same method failed once those non-physical terms were removed. A learned policy’s success is therefore only as meaningful as the physics it was trained on.

B. Rotor Angular Momentum as the Dominant Difficulty

Both approaches were shaped by the rotor. Earlier landing-first PPO runs held mean peak yaw rates of 1,625–1,651 °/s from 32.5M to 200M transitions. That is the momentum exchange predicted for their descent throttle swing (0.84 to 0.23). Open-loop checks confirm the 46.5 rad/s per unit coupling to within 5–10%, with no drift from swirl. The convex controller’s worst transient, a cold spool-up, yaws the body at up to 148 °/s regardless of the controller, because the rotor’s angular momentum comes from the airframe. Rotor inertia is an estimate, and the yaw coupling scales linearly with it. It should be measured, for example by a spin-down test, before either controller is trusted on hardware.

C. Limitations of the Convex Controller

The convex controller is not model-free. Its gains, thrust bounds and deadband compensation come from the simulator’s estimated vane, servo and motor models, and they must be re-identified on hardware. Lateral precision (0.05–0.40 m) is now set by a deliberately slow lateral loop. Its gains were chosen while the coning limit cycle was present, and higher gains landed within 0.04 m in the offline replica. The attitude loop is proportional-derivative only, so a center-of-mass offset leaves a steady attitude error (1.7° in the combined-disturbance mission). A 1 cm lateral offset can drift the vehicle off the pad during terminal descent. A plan takes 30–110 ms against a 33 ms control period, so real-time use requires a latency-compensating state predictor. The long route plus landing at the enforced 4 m/s took about 100 s, and a 100 s mission window was too short by a fraction of a second.

D. Comparison with Quadrotor RL

Compared to quadrotor RL approaches [2], [3], the TVC control problem presents unique challenges. Quadrotors

have independent, throttle-decoupled attitude authority through differential rotor speeds, whereas the TVC vehicle’s attitude control is fundamentally coupled to the thrust state. A quadrotor’s counter-rotating rotors also largely cancel each other’s angular momentum. Here, a single rotor exchanges momentum with the body at every throttle change. This coupling means that standard RL techniques developed for quadrotors—such as separate attitude and position control loops with free throttle exploration—do not directly transfer.

E. Sim-to-Real Considerations

The corrected simulation models battery voltage sag, finite jet momentum, full rotor reactions, servo backlash, sensor noise, wind and center-of-mass offsets. The remaining sources of sim-to-real gap include:

- Unmeasured 8S hardware. Thrust, shaft power, rotor inertia, motor lag and pack impedance are planning estimates for parts not yet purchased.
- Jet and vane model fidelity. Equal streamtube sharing, vane coefficients, residual swirl fraction and COP positions are estimates. Ground effect, recirculation and wake interactions are not modeled.
- Servo nonlinearities. The 1° deadband is an estimate that the deadband inverse relies on. Load-dependent speed and temperature drift are not captured.
- Mass properties. Mass and inertia come from the authored USD, not from a weighing or inertia test.
- Latency. Planner solve time and sensor-to-actuator delay are not yet simulated.

X. Conclusion and Future Work

This paper presented a GPU-accelerated Isaac Sim environment for a thrust-vector EDF VTOL vehicle. The physics was corrected for finite jet momentum, battery coupling, full rotor angular-momentum reactions with energy-conserving integration, and force application about the true center of mass. On that plant we compared model-free PPO with a convex-optimization guidance controller. The key findings are:

- 1) Convex guidance based on lossless convexification [21] controls this vehicle well once adapted to it. With a first-order-hold thrust profile, a thrust-rate bound from the vanes’ yaw authority, and a minimum-energy power-cone objective, it landed all seven in-envelope benchmark missions softly and on the pad. These included combined disturbances, a 50 m cold-rotor start and a waypoint route. It also flew a 100 s three-hover route and landed four further starts from about 100 m in wind at initial speeds of up to 30 m/s.
- 2) On the corrected plant, the convex controller substantially outperforms PPO: the best PPO policy reached 1.3% full-task success after 700M transitions and did not learn soft landing. A PID baseline also failed the soft-landing gate on the default mission.

- 3) The rotor-body angular-momentum exchange (46.5 rad/s of body yaw per unit change in rotor fraction) is the dominant control difficulty. It sets the admissible throttle rate for both controllers and couples PPO's throttle exploration to yaw.
- 4) PPO's landing success depended strongly on non-physical plant terms: 95.7% on a plant with unbounded vane forces and a tuned damper, and 0% once they were corrected. Physics verification must precede any claim of learned control.
- 5) The original findings on reward design still hold: initialization priors and terminal rewards that dominate integrated per-step costs are necessary to avoid degenerate PPO optima.

Future work will address the following directions:

- Like-for-like evaluation. Run the convex controller on PPO's randomized waypoint task and benchmark routes to obtain directly comparable success rates.
- Convex controller refinement. Re-tune the lateral loop now that the coning is removed, add a body-frame attitude integrator for center-of-mass offsets, and add a latency-compensating state predictor for real-time solves.
- Hybrid approaches. Investigate learning methods that use the convex planner explicitly, for example as a teacher or as a guidance layer, as part of the task definition.
- Gated Transformer-XL (GTrXL) policy. Replace the feed-forward MLP with a GTrXL architecture [16], [17] to provide temporal context for disturbances and long descents [19], [20]. This requires a sequence-aware PPO optimizer.
- Hardware identification and validation. Measure the 8S thrust and power maps, rotor inertia (spin-down test), servo deadband and vane forces, re-identify the controller gains, and validate on the physical testbed.

Data and Code Availability

The simulation environment, training scripts, configuration files, and evaluation logs supporting this work are publicly available at <https://github.com/Jacob19999/Transformer-rl-retro-propulsion>.

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